ActInf Livestream #022.0 ~ "Psychophysical identity and free energy"

Livestream | *1:30:03*

hello
and welcome everyone to actin flab this
is actin flab live stream
number palindromic 22.0 it's may 13th
2021 and this should be a really fun
discussion just to introduce ourselves
at the beginning i'm daniel i'm a
postdoctoral researcher
in california and i am joined by
oh i'm dean i'm from calgary i'm retired
and uh i'm i'm kind of new to this
active inference thing but i'm
trying to keep up i'm blue knight i'm an
independent research consultant based
out of new mexico
nice so thanks to both of you so much
for helping a lot on the slides and
giving insights and we're all going to
be sort of working through this together
because there's a lot of big ideas that
are going to come up
in this discussion and then we're going
to have the next two weeks
with alex kiefer the author and other
participants
on the dot one and dot two streams so
we're just gonna be surfacing
probably more questions than answers
today
welcome to actinflab we are a
participatory online lab
that is communicating learning and
practicing applied active inference
you can find more information at these
links this is a recorded in an archived

live stream so please provide us with
feedback so that we can improve on our
work
all backgrounds and perspectives are
welcome here and we'll be following good
live stream etiquette
we're preparing and setting the context
for the discussions on may 18th
and may 25th 2021 where
we'll be discussing the paper
psychophysical identity and free energy
which is a paper in 2020 by alex keefer
and just like the other dot zeros this
is a context and a background and just
an initial stab
it's not a review paper it's not
authoritative this is
us unpacking and exploring as we read
the paper
in preparation for the group discussions
and similar
to some of the other dot zeros we're
going to start out with
in the author's own words what are the
aims and claims
as well as abstract and roadmap and then
we're going to ask some big questions
we're going to talk about why it matters
which is hopefully a discussion that
includes everyone
and then we're going to meander through
some quotes from the paper
some formalisms some related reading
just a few ideas that we
actually brought in during our learning
process
to help connect the dots because the
paper is really like the tip of an

iceberg
and there's a long history of
information and thermodynamics
and that's even before you get to
everything with the free energy
principle
so we hope that this will be a
educational experience whether you're
familiar with active inference and just
learning about thermo
or vice versa or not familiar with
either
any overarching thoughts that either of
you would like to bring up before we
jump in
nope all right so
i'll read the first aim of the paper and
then either of you would be
welcome to give a thought the target
thesis of the paper
is the claim that biological systems
whose internal states come to encode
statistical models
as a result of spontaneous
self-organization in response to
environmental pressures
learn and make use of these models by
minimizing their physical free energies
the identity thesis which is that yellow
highlighted section
does not specify precisely which
approach to variational
inference is thereby implemented or the
minimum value the variational free
energy must take
it may be there may be for example that
biological systems run an algorithm
closer to contrast of divergence than

simulated annealing
so a lot of terms maybe first time
hearing it or not but what's something
that
draws either of you or made you curious
to read about the paper
well for me it was it was contrasted
with a contrastive piece it was
the idea that in order for this any
any kind of a modeling effort to work
you can't start out by knowing you have
to kind of
have a bit of a fuzzy idea before
there's anything that's edgy
cool and the sec blue
yeah so the second piece of the aims and
claims here and
very timely what is exciting about an
identity thesis in 2020 and we also hope
in 2021
and probably beyond is not so much its
opposition to cartesian dualism because
many people are that way today
but what does make it exciting well one
the fact the connection between mind and
body is drawn at the level of physics
undercutting even neuroscience as a
reduction base
so strength for some weakness for others
probably two
that the reasons for which and the
precise way in which
identity is realized can be discerned at
least in broad strokes on the basis of
current science
and three that the theory is accordingly
expressed in quantitative
terms so this is not your

run-of-the-mill anti-cartesian dualism
this is that 20 20 flavor and it has
more science now
so this for me was really awesome i mean
i am always looking for
um the expansion of the mind
brain duality to incorporate the body
um because i think that it's been
overlooked for
quite a long time and it's it's nice to
see that incorporation in this thesis
cool we'll definitely be able to return
to that
all right so let's turn to the abstract
it's three sentences long would either
of you like to read it
i can an approach to implementing
variational bayesian inference
in biological systems is considered
under which the thermodynamic
free energy of a system directly encodes
its variation of free energy uh
in the case of the brain this assumption
i like that
places constraints on the neuronal
encoding regenerative
and recognition densities in particular
requiring a stochastic
population code and some people may know
what that means but
i'm sure we'll parse that in a minute
and the resulting relationship between
thermodynamic
and variational free energies is
prefigured in mind brain identity
theses in philosophy and
in the gestalt hypothesis of
psychophysical isomorphism

and i didn't know anything about gestalt
before i read this paper
yep this paper it's like relating the
relations
because the little kernel down there at
the root is
mind brain identity theses and we'll
return to this but that's the
relationship between
mind and brain between the physical and
the mental
and then how is that as a kernel
the mind-brained identity thesis related
to
on one hand qualitative areas like the
gestalt theorizing which we'll talk
about
and then on the other hand more
quantitative areas
like the formalisms related to
information dynamics and thermodynamics
let's go to the roadmap so the roadmap
it's nice because first off there's only
a few stops on the roadmap
but there's a lot of freeway between
each stop
on the right are the sections that were
in the paper
and on the left is how alex wrote out in
a narrative way
the road map so we kind of have the turn
by turn on the right side
and then we have a little bit of the
expert giving you a guide book on
the left and it starts with an
introduction and goes straight to
formal equivalences between what well
between free energy

and a lot of other things a transparent
code a little bit of an oxymoron
there i thought codes were supposed to
be oblique
but what is a transparent code and
eventually how does it
lead to this free energy identity thesis
being proposed in 3.2
and then discussed more in section 4.
then there's a conclusion so it's a
pretty dense paper and what
did anyone think about just the road map
or just getting from a to z
can i speak to that for a sec please
yeah
so the first reading of this when i saw
transparent code i wondered about that
as well
um a lot because i didn't even see it as
oxymoronic i just wanted to know
how that was going to get unpacked
and then the other thing i found really
interesting on first reading
and and both of those sections three and
four
changed by the time i read through the
paper a third time
but then the whole idea of isomorphism
to identity
um i came around to the idea that i
think
i think he was talking a little bit
about the ability to be an identifier
as one of those measures of the sort of
quantitative measure
of what an identity could look like he
doesn't really get into he still
throws possible in there quite often in

the paper which i think is really
a good thing to do because he doesn't
try to
doesn't try to uh bring it down to
absolutes which i think is
is smart given as you said the density
of the
the stuff he's trying to look at
cool blue anything otherwise we'll go to
the
big questions so um other than
it was a really pretty rocky road to get
from a to z
um and it's uh you know i i
just kind of had no idea really where
this was going so it was nice to see it
evolve um as it unfolded i just always
keep in mind
the paper's finite i will get to the end
if i just crunch line by line
and somebody who knows way more than me
on this topic
thought way more than me on the paper so
let's just
trust the author not in every single
claim but at least in the structure
and go from there well why does it
matter
why are we spending our regime of
attention
reading this paper clearly multiple
times each
and digesting it and turning it around
and having discussions on a chat forum
what is bringing us to the table from
very different backgrounds and hopefully
you as well
to think about and be curious about this

paper
um the three things we wrote down here
and then either of you feel free to
unpack one or any
why does it matter well continuing
existence depends on efficient energy
use
it's important to understand what life
is as well as how we model it
and it's interesting to think about
information identity
and physics what brought any of you to
the table
well so just to comment on um the
distinction between life and how we
model it i mean i think
this is very relevant to the discussions
that we've been having on realism and
instrumentalism
and really a fundamental way to
clarify that distinction i think which
was nice
cool and the big questions the paper
which are sort of
a little bit drawing from quotes from
alex other times us
linking the questions together one is
how might models of psychology
action and perception such as gestalt
but basically just
psychological theories that might have
no math
involved at the first pass how could
those
math-free ideas be connected to formal
equations
potentially those related to certain
kinds of statistical inference

so how are we going to connect
psychology to equations
big question two after realizing some of
these resonances
between certain types of statistical
inference
and information and thermodynamics as
well as potentially
mind and the brain is there any way to
bring that all together you know we have
apples and oranges but maybe we can make
a still life that has apples and oranges
and they're each in their own place and
it kind of makes sense
and then the third point here is the
biggest question perhaps which is
where does identity come from or what is
identity
how would we measure that or
characterize it or determine it
and i think um one hint
perhaps from dean was to start with
identity formation
uh what it what it is not so
then there was a quote there about
comparing with uh
neuronal networks and computers
any other thoughts here before we jump
into keywords so
this quote like the the difference
between living systems and computers i
think like
it would have been nice to see that up
front uh i
i kind of had no idea where the paper
was
going and like that it was gonna come to
this like really

um solid kernel of meaning
right and and so it would have been nice
to have some hint of that like
foreshadowing maybe like an introduction
or the abstract but i was like
had no idea where this was going and
actually pleasantly surprised like
wrapping a gift wrapped in
you know 10 different boxes at the end
it went better than expected
you know in active inference what could
we want other than that
but it does say that it provides a
criterion for distinguishing minds from
relatively crude simulations
like blue's alive and was just speaking
but if it was just the voice being
played from a speaker
is that going to be counted as alive
because it's making noise as well
well if not we want something richer
than just making noise what is that
going to be
how are we going to get there we're
going to start with some of the keywords
and just like there was only three
sentences in the abstract and in the
final paragraph
there was only three given keywords and
so
rather than do a keyword driven intro
and then
walk through the paper we're gonna kind
of do a little bit of both
we're gonna give some background
information and some frame setting
on the paper and then we're going to
walk through the paper and where we need

to introduce a formalism
we'll just kind of unpack it there and
share resources
okay so earlier we were talking about
how it's kind of relating
different relationships and this slide a
little bit abstractly
a little bit provocatively a little bit
ambiguously
is relating those relationships so the
top
yeah yeah okay
because of the twist and i i i think
when i saw it i mean i
didn't anticipate ants which is which is
cool too
but that but the fact that there's
there's a twist
in it i actually think is
is what we're talking about it's i
i think i think that will i think that
will again
surface as we go deeper into today's
conversation but actually
i was like whoa very cool cool
so what we were trying to get at here is
to
link the topics and then see how those
topic linkages
are linked so the mind and the brain are
these two
nodes on top and the question is how are
the mind and the brain related
and you can check out the mind brain
identity thesis
to learn more about what a few previous
perspectives have been
but one can imagine there's all kinds of

takes they're the same thing they're
totally different they're a little bit
related
so that's how the mind and the brain are
linked and then
on the bottom we have these two nodes
one of them is
actual thermophysics so candles
actually burning balls actually rolling
to the bottom of a hill
and then on the right side we have human
models
resembling patterns that we see in
thermophysics
and the question of the linkage here is
how are just like how are the mind and
the brain related
how are actual thermodynamics so instead
of skin in the game atp in the game
actual thermo related to maybe some
equations in other domains that make us
think about information dynamics so
those are
the level one linkages first between
mind and brain and second between
actual thermo and models that are akin
to thermo
and then this next slide is a little bit
of a
multi-play but it's saying maybe by
juxtaposing these relationships
also with a nod to gestalt and illusions
with the arrow illusion maybe by us
as the observer juxtaposing these
relationships
we gain new insight and maybe fep
which is sort of the little ants
crawling all around the sides

maybe the fep is that permeable membrane
or it's the
framework that's strong enough for us to
build within
and sub-specify but it's also permeable
in that it allows us to be open to new
ideas coming in
and the psychophysical identity thesis
is kind of
smack dab in the middle it's relating
these relationships
so if there were four things that get
brought together i think it would be
these four
mind and brain and then actual thermo
and models of thermo
and the way to link them is to first
make the pairs and connect the pairs and
then see how those linkages are
contrasting or similar
danny would you say that part of this
too though is the fact that they are
parallel and apart i mean we build a
relationship
we've already sort of confirmed that but
is it
is it is it a connection and i'm not i'm
not saying that in a questioning way
it's just i think it's a legitimate
question i'm not sure
that the way that we quantify that
confirms it or not but maybe i'm
grasping at something that isn't there i
agree with that
that this relationship exists but i'm
not sure
if it's a connection but that's just me
wondering

perhaps it's reflective of the
instrumentalism versus realism
debate right like so thermo physics is
um a model
and the human system is like an alive
system with you know some kind of
concrete accident existential
purpose right so we assume yeah
it's almost like real on the left side
and then more like an instrument
on the right side because the brain does
have
thermodynamic constraints it generates
heat
it takes in glucose it's doing
metabolism if you cool it down
it stops so that's like a candle and
then
what is the mind especially if the mind
is kind of higher order
and reflexively making a model of itself
in an interesting way
but if the brain like a physical energy
consuming
object is like a candle that's this left
side
people might draw a link between those
two draw an edge between the brain and
actual thermo and we're gonna get to
that in just a few slides
and then the question is well what's the
relationship then between these
thermo-inspired models on the bottom
right
and the mind is it that the brain is
actually minimizing free energy or is it
that we can model it as minimizing free
energy

what is happening here let's go to the
psychophysical isomorphism
i'll play this video and um blue what
made you interested in this video so i i
saw the term
psychophysical isomorphism in the
abstract and i was like
what is that right it's the the uh
gestalt hypothesis of psychophysical
isomorphism and i was like those are
really big words
um i just don't have the background so i
wanted to look up the concept
and it's it comes from the expression
that
um the properties of the mind and the
consciousness are a direct
consequence of the electrochemical
interactions within
the physical brain and so this um
this is the the five phenomenon the
video did it work
it's not playing on my on my end so this
five phenomenon is an example
of where the patterning of a stimulus
and the activity in the brain is similar
and that's why it makes this like
illusion of motion
cool so it's almost like we're
connecting the dots
but here it's shown with negative space
and that's
a entry point into thinking about
gestalt more broadly or is there
anything else you want to add on this
slide
okay that's it what is gestalt
well people live their life teaching it

so we're not going to cover it in one
slide but we can cover some of the ways
in which it was used
in the paper as well as just a few
little tidbits which kind of
on our active inference foragers path
were cool
so first just as a design principle on
the slide
you visually and this is a culturally
uh scaffolded process separate
for example if you're not colorblind the
red
shape out from the black text and so
it's almost like there's this
organization that arises when we're
viewing material
and that turns out to be patterns that
we can study
like reification on the left side here
we do see
quote a triangle like a white triangle
in a even though it's the negative space
of these other
pac-man type shapes and so gestalt is
this big
area of theory so check out
the word itself terms like figure and
ground
and all these related fields and what
are we getting at
that's bigger than all of these examples
what does reification of shapes
and multi-stability like illusions
have to do with this notion of
invariance
like the idea that by looking at
different angles of a shape we can

recognize that an object is invariant
even though computers often have trouble
with that so it's almost like
if you're working towards grasping it or
you're seeing that there's something
bigger behind these examples
that's kind of gestalt it's hinting at
the whole
and so it's very related to holism and
relational thinking
but we can dig one more level into
gestalt
and just look at how alex used it
and in the paper it says um
this work the paper that we're reading
and other work by carl fristen and
collaborators
confirms a systematic link between
variational free energy
even if it's your first time hearing it
it's all good variational free energy
and thermodynamic potential
energy interestingly this view is
presaged
not only by the identity theorists in
philosophy
but also quite precisely by the gestalt
psychologists
who suppose that the perceptual
phenomena as subjectively experienced
had structures isomorphic same shape
to their underlying physiological
correlates
and does that mean when you're looking
at a triangle that there's another
triangle that's the same
exact thing in your head well not the
same exact thing but maybe something

that's
structurally isomorphic or the same
shape
and so to unpack that and to also
evaluate the author's claim
is it really true that this area of
science that we think is super awesome
and advanced in 2021
was it really true that it was presaged
so we can go to these citations
like this 1935 citation and just copying
out the contents and doing a little
color coding
look at the titles of the sections
chapter two has to do with behavior
and sets up the problem but chapter
three through seven
it's about the environmental field the
field of affordances
this is what we've been talking about
with a lot of our different discussions
and then action memory learning and at
the end a little bit of a closing note
on even society
and personality and probably what it
means to be a person and an
individual so we are seeing a lot of
these terms
which should really bring us to slow
down and think about
how we can tie different knowledge
traditions and perspectives
together in the way that alex has done
and hopefully in a way that we can also
do when we come together to discuss
any thoughts on gestalt
okay now we're gonna switch gears
from the ecological field and psychology

to thinking a little bit more
quantitatively
about statistical inference and
metabolism
alex wrote that the argument at hand is
meant to support a slightly less
abstract isomorphism or identity
directly between variational and
thermodynamic free energy descriptions
this argument is consistent with a
detailed treatment of metabolic
efficiency
of variational inference given in 45
which is a
san gupta at all paper with carl fristen
in 2013
so for those who are interested in
active inference check out where active
inference was in 2013
and think about how what has happened
since then changes our understanding
but just to pull one figure from this
paper which is thinking about energy use
in nervous systems
um it's showing how the sensory
environments whatever's out there
is going to get encoded this is where
that transparent code is going to come
into play
into a format that allows
control gain control of the sensory
input as well as action
control and it has to be done in a way
that's thermodynamically
viable so you'll often hear things like
the brain processes this much
information or it uses that much energy
as if it were just like some sort of

desktop computer
but it's pretty clear from just a first
pass understanding
that it's not just doing what my video
card
is doing vision isn't just the ram
going back to the video card and back
and forth with a processor
so how is it that brains and cells and
all kinds of biological systems
are able to act effectively when it's
pretty clear that they're not doing it
like
our heat producing computers are doing
it
so this is sort of where we're looking
at how do biological
organisms do good inference on the very
complicated
sensory environments and represents
all of these wild stimuli in a way that
allows for effective control
of the inputs and do that without
overheating or requiring just a ton of
energy
let's go to free energy
want to start here blue or i can start
yep
free energy in the context of
variational bayesian inference
is a functional of a probability
distribution or density q
used to approximate the in practice
typically intractable
joint posterior distribution p of h
v of data v together with the unobserved
causes of those data h
under a statistical model whose

parameters may also be
unknown um and
sorry go ahead oh just that q is the one
that we can control and q is sort of our
guy
and then the one that we can't get at is
the
actual distribution p of
two different things the data which is
what we definitely
have so that's good but then the
unobserved causes of the data
h and sometimes h is hypotheses like
i hypothesize that it's raining and so
what are the data that you get well
maybe it's visual
or maybe it's tactile data but you're
going to be able to control
your estimate of whether it's raining q
and then what you're trying to get at is
that total distribution p
part of which you can get at with h and
v
so we're going to be trying to talk
about a way to bring what we can control
our internal generative model and align
that distribution through optimization
to really complicated distributions in
the world that might have many more
degrees of freedom or a lot more
variability than we could possibly
handle
and another nice quote on the bottom and
yellow is
free energy captures the discrepancy
between the organism's generative model
of the world and the current
environmental conditions

so that's kind of cool and we've seen
that come into play
in the context of the predictive
processing or the hierarchical
predictive processing framework which is
shown here from
ramstead at all answering schrodinger's
question it's a figure
that's showing how there's sort of a
bottom-up which is sometimes associated
with sensory input
and a top-down sometimes associated with
priors about the world
these two interacting streams and
they're
in a multi-level way so each level is
only getting passed up or down
one level and what's going up
are these sensory inputs and unaccounted
for
errors and what's coming down is
expectations
so if the bottom up is exactly what you
expect it's almost like the system is
resting
so in other words if the model is
perfect given the visual coming data
coming up
everything is very chill and things are
not
chill to the extent that the bottom up
information
is surprising so what is getting passed
up
and under the free energy principle
maybe it's free energy
but what is free energy
another way to look at this question

about the relationship between
the observations in the world which are
called data
in many cases and the generative model
of the world
is with uh the paper that we discussed
in active number six
tale of two densities and
variables and parameters they're used
often loosely because when you're
dealing with programs
everything's a variable everything's a
parameter so
it can be a little bit confusing to
which kind is which
but the data parameters are the
observables and the recognition model
in bayesian statistics is the model that
recognizes the data and forms hyper
parameters about the data
for example um the hyper parameter might
be like the location of an object
and the data might be the actual retinal
cells giving
raw input and so the recognition model
is going from that retinal input
to a hyper parameter of where's the
object located
and then that can be used in an
invertible
way with a generative model that goes
from okay given where it is
in the estimated location what kind of
data would be generated from that
so this is a bayesian model fitting
approach that's related to expectation
maximization where you take what you
have as far as data

and you update what you think about the
generative process of the data
and then you crank it the other way and
you ask given what we know about the
world
what kind of data might it generate and
tale of two densities
contrasted that sort of bayesian
approach with a lot of the theory of the
inactive
researchers who are interested in this
relationship between agents in the world
and seeing that incoming data as
analogous to perception
and then the outgoing um generative
model
as generative model of action so that's
what we discussed
in tale of two densities dean yeah i
wasn't there for that but i'm just going
to ask the two of you
um i really like the the diagram but one
of the things i was curious about is
i've
had lots of conversations with people
who have taken an activist stance or an
act
an activist perspective on things i'm
wondering
from from you guys perspective is this
ability to ask and then what happens is
it how is it fundamentally different
on either side of that um that that
diagram fundamentally different
so you're asking the ability to ask and
then what happens
so the ability to ask a counterfactual
yeah

how is it different between the
statistical basis
and the here i am in reality basis
because i've asked this question and
every time i ask it of people on either
side of that
of that diagram they kind of go on
i'm not really sure do you have any
thoughts blue
yeah i um you know i i think
that this paper kind of really tries to
get at it actually like getting at
the the what's different um you know in
maybe not this inactivist sense uh but i
yeah i really don't have the um the
background but i think
you know a lot of times there are
layers that seem like they should
overlap but maybe don't have a direct
curl it right and i think that that's
what's happening here and i think that
that's what alex tries to address in his
paper
in like a thermodynamic free energy
statistical
thermodynamic kind of way right
fair i'll give another thought so and
then what happens
in the context of bayesian statistics
it's like a
question about counterfactuals so if
you're asking and then what happened
differently with the data
well from the point of view of the hyper
parameters either the data would be
better fitting or not
if you ask and then what happens to the
hyper parameter well it might generate a

different kind of data
what would happen if the sun were in a
different location well the shadow would
be different
so as opposed to that sort of cut and
dry sense
of counter factual either on the data or
the generative side
in the inactive frame perhaps
and then what is more of an invitation
to play
so just at a very first pass the
inactivist perspective
gives more of a affordance for
negotiating that counter factual
as um play rather than just
coercing it into a statistical crank
it's like well what happens if we jam
this gears like well then the wheel is
going to stop turning
next question but inactivism is a little
bit more playful with that
blue so i didn't see this
in terms of counter factual i i was
looking at it in terms of
a predictive standpoint right and then
what happens is
in my mind a prediction and so making a
statistical prediction versus making
like a
prediction on the side of the organism
or from like an inactive
standpoint to me like i don't know i
mean i still think like
it just you predict and then either
you're right or you're wrong and then
you update your model accordingly
yeah dean yeah so

because i've sort of come at this from
from the
wayfaring and the wayfinding of
how do i orient myself to this stuff i
wondered i've wondered i don't even have
the
confidence to sort of put this out there
and and stand by but i've wondered if
part of it is
the statistical gives us a bit of an
allocentric a bit of a detached
kind of back to what both of you said
and daniel
you know there is a personal piece an
egocentric part on the inaction side
that doesn't necessarily exist
on the allocentric side but that doesn't
make one one or the other
more or less real but i think it's
whether you can look down on something
and see it in that kind of a
relationship because we're talking about
relationship or relationships
or whether you're seeing it from the
ecocentric oh
that that snowball's now going to hit me
duck right
um so i don't know if that's the
explanation that that
kiefer was going for here but um
yeah i i think it's i think it's part of
the conversation if we're going to
quantify this
so perhaps the difference here really is
and then what happens
in my model and then what happens to me
very nice very nice it has to do with
with agency and

there's an eye in that inactive side
there's what happens to me and what
happens to my model or how would i see
it differently
versus given this chess board
how would it just play out which is a
little bit more like the impersonal
bayesian
so pretty cool and the reason why we
showed this slide was first off to
highlight that even when we're talking
about the hard
the soft is there and maybe even vice
versa
and also because this approximating
distribution
that the free energy is being discussed
in the context of
is the recognition model so that's why
we're showing this
mapping okay what is variational
inference
well it's something that i'm sure we all
had to do a little bit of a
dig to find out about and early in my
digging
last week i came across a really helpful
pdf
and i started with this quote from what
alex wrote
and then just started searching and the
quote that alex wrote was
that variational methods formalize
statistical inference and learning
in terms of the maximization of a lower
bound on model evidence
called the negative variational free
energy he describes them

as state-of-the-art approaches and also
notes that the variational methods have
a long history in
unsupervised learning so cool it sounds
state-of-the-art but there's a long
history
and maybe it's a tractable way to get at
some hard problems
well what is it in this pdf started
pretty early on said if you haven't seen
jensen's inequality
spend 15 minutes to learn about it but
again
this person has thought about how to
teach variational inference maybe it's
worthwhile
thinking about jensen's inequality and
one way of phrasing jensen's inequality
is a generalization of the statement
that the secant line of a convex
function
so kind of like a chord on a circle like
a cut through a circle
lies above the graph of the function
so it sounds like it might be a little
bit obvious graphically and indeed
there is a proof without words so that's
just flipping to this next
slide so yes it's kind of like if
there's a bowl
any slice that you take through the bowl
it's gonna be able to have something
roll underneath
the slice that's really important
and pulling a few quotes back from
the variational inference by blay pdf
what is written is that we can't
minimize the kl divergence

exactly maybe we'll talk more about kl
later
but we can't minimize the divergence
between um
the data we're getting in and the
generative model we want to minimize our
divergence so just we'll think about it
at that level for now
it's hard to minimize that directly but
we can minimize a function
that is equal to it up to a constant
this is the evidence lower bound or the
elbow
so it's almost like i can't tell you
where the bottom of that bowl
is but if i can take a cut through the
bowl
then i know that it has to be lower than
that but if i cut as low as i can to the
bottom of the bowl
maybe i'm going to be just
infinitesimally cutting above the bottom
so if i have a really easy way to figure
out how that cut gets minimized then
maybe we'll do all right maybe we don't
need to find out the true single bottom
because we'll just make cuts that are
lower and lower and we'll go from there
and it turns out that jensen's
inequality really comes into play
and also part of the trick is that
there's a
choice of the investigator for
which functions to use and it turns out
that
some functions it's easy to calculate
them one way
but really hard to go the other way and

there's times where you want that like
when you're doing cryptography
but there's other times where you want
functions that are
very very easy to go back and forth like
as easy to go
from from the um input to the output
it maps uniquely and quickly back and so
it turns out
this loop thinking kind of primes the
pump but it turns out if you choose
mathematical functions where it's really
easy to go from input to output and back
and you combine it with this idea that
you can take a slice
through a curve and find
a tractable way to estimate basically
the uh lower bound on the evidence it's
gonna be possible
to set up some optimization problems so
that we can solve them
straightforwardly and so in other words
highlighted here minimizing the kl
divergence
is the same as maximizing the elbow so
maximizing the evidence lower bound
means like we're pushing
up the bottom bar so we're getting more
and more evidence we're getting better
optimization
as we minimize the divergence between
the data that we're getting
and our generative model but we picked a
generative model that we can play really
flexibly with
any thoughts or we'll continue on
just that it goes back to solving the
problem once you know the answer

solving the process of solving the
problem once you know the answer becomes
like very constrained and simple nice
and i just described this as then if
then which is exactly the same thing
then you have you have something that
you already are aware of
the if the unknown space below that line
and then
you resurface to a new place and and
below that line on that next
slide is what i would describe as
learning's unholy triad
the triangle under there but i'll let
you talk about that
no it's it's really interesting because
um
outside of the cord you know outside of
that line
you have no guarantees and in fact often
you increase the precision of your
guarantee
within the frame at the cost
of losing the generality beyond it
doesn't make any generalizations
so the metaphor that i don't know
somebody with more math please
correct or help us make this richer but
it's kind of like
variational inference is doing inference
on rails
so the train operator can have a really
complicated generative model of the
world like what time should i be home
for dinner
and what's my favorite color uh could be
wild sensory input
so you can have a really rich system but

the oh the operator can tell from the
local speed of the train
that the train is going in the right
direction and maybe even how fast it's
going in the right direction
given the structure of the problem go to
the desired location
now there's other times where you don't
want to go as fast as possible but just
taking it at a first pass going from a
to b we want to structure the problem so
that we're on rails
we know that all we have to do no matter
how much data we have in our generative
model
we want to have like a forward and break
and so variational inference helps us
structure the problem so it's more like
being a train
on a track with a destination in mind
which is this minimization of the
divergence which we will get to
as opposed to sort of this like hill um
characterizing and just mapping all the
hills that's like a map maker
but the train conductor is not map
making they have
exactly their tools at hand and then
they're just going as fast as they can
in the right direction and
this was just a technical slide from
the paper any remarks on this blue
otherwise i think we could yeah okay so
this is for those who um
want to see the technical details read
the paper
but this is equation 2.1 and it relates
some of these terms that we're talking

about but
we're just exploring today this was
another useful
blog post and it
plays into the fact that the kl
divergence
is asymmetric so the divergence between
q and p is not necessarily the same as p
to q and so it's almost like there's two
different um ways that we might optimize
it's kind of like finding
a local minim uh local maxima versus a
more
global maxima but q
is a constrained type of distribution so
for example
here we might be picking q to be a
gaussian so we know that q is going to
be a bell curve
and we're only going to be trying to
maximize one thing about q
which is its mean and its variance as
well
in the easy case what the author of this
blog post
writes as easy is just like find me
the highest point in p and then
place q right on top of the highest
point in p
so that gets you to a local
max it finds a local the global maxima
and it just lays down
your average there
but it turns out that there's a hard
direction as well
which is actually given the total
distribution of p
i want the best overall fitting queue

and the author writes well why is this
hard to compute
well for most interesting models we
can't compute a complex model
for example the real world or really
complicated graphs or very complicated
bayesian posteriors
because remember a lot of this
variational inference
rested on the fact that we were choosing
distributions
that were easily reversible however
when we're optimizing functions where
there isn't a really easy reversible way
it's harder to do inference
and they also write down here on the
bottom
the optimization problem is convex so
curved a jensen's inequality
when q is an exponential family i.e for
any p
looking at this bottom graph the
optimization problem is easy
so in other words if we pick q to be the
right kind of distribution
then it'll be smooth sailing no matter
what p looks like but if we want q to be
the same type of distribution as p
and we don't know what kind of
distribution p is we're kind of
setting up the problem to fail however
if we say hey whatever kind of problem p
is
whatever is generating p we're going to
make
q part of the exponential family then it
becomes easy
and so you can think of maximum

likelihood estimation as
a method which minimizes the kl
divergence based upon sampling p
but in this case p is the true data
distribution rather than
a model of the generating function dean
yeah it was interesting because when i
first looked at that i wondered
is this is this a good way of
quantifying the difference between the
guy in the train who's looking at the
world through deontological eyes and the
another trained conductor who's looking
at it to utilitarianize
because that that actually exists in
those those graphs
it's kind of interesting
i'm thinking of who who those different
kinds of people would be or what they
might be well you know
yeah which one is if i'm gonna run over
somebody on the tracks do i run over
uh five do i run over five people who i
don't know
or do i run over one person who i do
versus i can't do anything that i
wouldn't have
something be done the equivalent to me
but anyway i just think it was
interesting
that the qual the philosophical
qualification ended up being
visible in the representation of those
of those
graphs there
cool and here is how alex
framed it which is that f the free
energy

is useful as an optimization target so
just a little reminder why are we going
into all these details about statistics
and about the kl divergence and the
variational inference
well because we're talking about
variational free energy and doing
optimization on variational free energy
so that's why we had to pull back to
asking what variational
means and then when we think about where
the energy comes in it will return
so alex writes again that f is useful as
an optimization target for several
reasons that have been widely discussed
and
maybe those reasons are obvious to some
or some people
think that it's uh not enough of a good
reason for f to be used
but there are reasons to choose f as a
function that you want to minimize
2.2 gets at that and it even uses the q
and the p
so if we look at 2.2 relative to 2.1
in 2.1 see how p is on top and then q is
on bottom
and then here q is on top and p is on
bottom and there's this plus
f if you want to unpack the equations
please do so and help teach to us but
just note that
things are flipping and it's being
mirrored by what we saw here
where on the top one the easy one cues
on top
right and then on the hard one q's on
the bottom

so also using that phv notation
and then um what
does that mean so let's just say that
free energy
is the thing that we're going to be
optimizing on
and variational free energy has a
certain structure
that's related to variational inference
that means that maybe
if we're trying to minimize variational
free energy that's going to be like the
right optimization
path okay so we're going to be doing
variational free energy minimization to
do
model fitting where does real
energy come into play where does
thermodynamics come into play
and it's right here as is often remarked
in discussions of variational inference
so eg in the statistics literature
f free energy has almost precisely the
form of a negative
helmholtz free energy from statistical
mechanics in fact that's why we call it
free energy because it has a really
similar form
and that free energy term or the energy
term let's just call it
f of t um
it's uh has two parts to it the first
term on the right
is the expectation so that's e is for
expectation here
of the energy stored in the system at a
given temperature
due to internal states so really

complicated molecules with a lot of
uh order like dna they have a lot of
internal energy locked up in their
structure
and then the the right term ts is the
temperature times the entropy
so that's what the term looks like for
uh
thermodynamics and that's how you can
look at whether an equation is going to
be like endothermic or exothermic
like if it creates a lot of disorder in
the environment it can locally order
some stuff but you know you can't
make or lose energy and so to get
to this more chemical phrasing of
f from variational inference alex writes
we rewrite equation 2 1
using an energy term and so we get this
tooth equation 2 3 which is analogous
to this more statistical physics
equation where we have basically vfe
consists of
the energy of the internal configuration
minus the temperature times the entropy
so part of the question is going to be
okay cool it looks similar
might those things be isomorphic might
they be the same
that's the thesis that's the question is
it
actually that way or is it just that we
can
think of it similarly and so this is
where we got to 26
bringing back the train choo choo could
we take well-structured optimization
problems

and connect them by resonance in the
structure of the equations
to the ways that downhill trains
navigate
or burning candles or if you put a ball
into a bowl
it goes to the bottom every time and it
only goes to the bottom
so could optimization variational
optimization
not just be well structured not just
have the gas and the break
so that we know that we're going in the
right direction but could we be
thermodynamically pulled in the right
direction
that would be like the candle where the
candle always burns
if you think where its higher state is
with a candle
and then the lower energy state is it's
burnt and dissipated
it always happens given the activation
energy
and if there were an isomorphism between
the structuring
of the statistical problem and the
actual thermodynamics in the world
that would be pretty interesting
so analogies are always partial i know
dean's probably going to have a ton to
say about analogies here
but this is what is written by alex
if the connection between statistical
mechanics and statistical modeling by
the brain were merely one of analogy
it would be surprising why to find that
all the terms in the helmholtz free

energy
play useful in interlocking
representational roles
this coincidence between physical and
representational descriptions is to be
expected however
if free energy simply measures how much
useful
representational work can be done by the
internal elements of a system
where work has physical meaning
in the remainder of section 2 i consider
how the physical interpretation of each
term in the helmholtz free energy
can be related to a corresponding facet
of the optimization process
so it's like we have a dna molecule and
there's the energy that's stored in the
configuration of the bonds that's the
internal energy
and then there's the temperature
dependent term
which has to do with the dna staking
around and moving around in the
environment
so now when we compare that to
statistical models it's almost like the
internal energy
is the complexity not of the dna
molecule but
the complexity of the model and then
that wiggle room
is related to potentially the accuracy
or the variability of the model
but we'll see
anything to add here on the candle
okay just focus that the focus is on the
function i think

the representation as function
yep that's really interesting so i hope
alex can clarify in a lot of these math
points
and also on the representational work
side
so here's another piece that came up
during
preparing for this uh gibbs free energy
is g and the difference between the
reactants and the products of a chemical
reaction
is often referred to as delta g
and when the delta g is significantly
low like it's a downhill then if you can
get over the activation energy
with just thermal energy or with an
enzyme lowering the barrier or whatever
it will flow downhill so if the delta g
is negative the reaction proceeds
as surely as water flows downhill or
in hill should i say because up and down
or not real
and so here's three different cases that
are being shown
so here is the downhill case where the
reactants have a higher
g than the products the
reaction proceeds but not all the way
sometimes it's functionally it proceeds
to completion but actually all reactions
have this k
value where it's somewhere in between
because at some point you'll have like
99 of the product
and then the backwards reaction is 50
50.
so just like if you have a test that's

99 accurate but the disease is really
rare
you might do the bayesian statistics and
find that actually contingent on a
positive test
it's actually not likely that the person
has it because there's so many people
who don't have it who get false
positives
that's the bayesian case so in the
chemical case it's kind of like
even if the reaction is favored 20 to 1
well
once it's 95 done then
for every one of those that goes forward
20 come back
so it re-equilibrates at 95 to 5 if
the overall difference is 20 to 1 and
then on the right side on the top
if the products have a higher delta g
they have more internal energy
then you need to add energy to make it
happen
and then on the bottom left is the
special case where the reactants and the
products have the same
gibbs energy so for example like two
isomers
of a molecule and in that case it's like
a coin flip
it's as uh energetic for the reactants
and the products
and so it ends up just jostling back and
forth and getting completely mixed
so those are the little bit of the
thermal intuitions
that people might have heard related to
kinetics and thermodynamics of chemical

reactions but let's talk about
inference and models what does it look
like to have a model that's going
downhill on an energy landscape
or a mod two models that are as likely
as each other so it's kind of like a
coin flip
between them so that's just kind of
again getting at this
potential um certainly helpful but
potentially even
physically real isomorphism between
model inference and thermodynamics
cool blue what could you
tell us on this slide so i think these
this is uh stuff that alex put in here
talking about
the role of internal energy and so these
two equations of the helmholtz helmholtz
free energy you talked about already
first
so where u is the internal energy the
top equation is a equals u delta s
and so u is the internal energy
combining the kinetic and potential
energy
total in the system and then the t
is the temperature and then the s is
entropy and
he writes an intuitive explanation of
the minus
ts term is that insofar as the
properties of the particles in the
system are uncertain
their kinetic energy constitutes
irregular motion
so the impact on the free energy of
entropy a measure of the irregularity

is scaled by temperature roughly average
molecular kinetic energy
um i thought that was a neat explanation
and then the boltzmann distribution is
this bottom probability and it's the
probability of
a state at a given temperature in
relation to
the energy in a system and he says here
low energies are associated with good or
desired
configurations of the system which are
often interpretable as assigning
high probability to what they represent
and the analogy to physics is closer
that the energy can be directly related
to the probability of the internal state
of the network occurring
cool so how exactly are these different
flavors of thermodynamics and math
related
alex teach us and and
help us understand where other
researchers can
contribute how about on 30
blue what was interesting to you about
this i threw these in
because these are both different kind of
neural network
setups and so like i'll start with the
bottom one
the boltzmann machine isn't really like
practical or
or used a lot um to solve problems
because
it's really takes a lot of energy as you
can see like the connections in the
network there's a lot

it's like a fully connected network but
they use this restricted boltzmann
machine
to kind of solve problems instead it's a
it's a neural network architecture
and it's simpler than the helmholtz
machine which is above it it's just
divided into
the visible units in green which is like
you can see the data going in
and then the hidden layers which are the
blue nodes
in the network um and it's uh
anyway it's just uses gibbs sampling to
recognize and generate inputs
um not like what goes on in the
helmholtz which is like the
there's in the helmholtz model which is
the top one there's this recognition
layer and a generative layer and so
there's the hidden layers are behind it
so
when you see the visible layer it's this
is the input in the recognition
model and then in the generative model
this is the output and so i think that
it's really kind of cool that there's
this recognition layer and the
generative layer
because it just relates um so closely to
active inference and
and how that's um all done and
uh there can be multiple layers so you
can have like a deeper
um you know structure than this and also
these helmholtz machines
are i think alex talks about in the
paper he doesn't show it

um or maybe say it exclusive explicitly
but the helmholtz machines
literally are doing sleep wake sleep
wake and so how you train
training the model is that like the the
generative tr
or the generative traces the recognition
like by definition and so it's you know
constantly chasing each other
versus the ultimate machine that's just
all kind of kind of happening at once
thanks that's really interesting also
these names like boltzmann and gibbs
it's almost like there's equations that
are named after them
in thermodynamics like gibbs free energy
but also this is an inference machine
the boltzmann machine is an inference
machine
and i i just looked it up uh some may
find this interesting
some may not but this is the wikipedia
page for gibbs
sampling what is gibbs sampling gibbs
sampling
is a markov chain monte carlo algorithm
for obtaining a sequence of observations
which are approximated
from a specific multivariate probability
distribution
when direct sampling is difficult so
that comes into play
in blue and i's area of genomics or in
phylogenomics where we don't have the
generative model
but we have a ton of data and part of
the way that we fit
bayesian models in phylogenomics is by

using these sampling approaches
and so it's kind of like sampling but
that stochastic way that we're sampling
where like you kind of
go in the direction of informative
samples but
also you want to sample broadly from a
distribution that you don't know the
real underlying truth but you just have
a ton of data
that is just very interesting to see how
it comes back
here so thermo and info have been
presaged for a long time that's what
alex does an awesome job of
bringing back some of the literature
from
okay what is temperature
so there's temperature in the physical
world which is related
to the speed of molecules
and just before we quote alex on the
left just
a reminder that even at a given
temperature
that the speed distribution of different
molecules is really different
because they have different masses so
what's conserved at a given temperature
like zero degrees celsius or something
is actually an energy value per molecule
and lighter molecules are moving faster
than larger molecules
and what alex writes is that the meaning
of temperature within statistical
modeling
so that's what it means in physics
statistical physics

but in statistical modeling is
illustrated in the example of simulated
annealing
where increasing the temperature
increases the variance of the
distribution over configurations of the
system
and lowering it collapses the
distribution to a small
range of states near the ground state so
like when you have dna
the lowest energy state is when the two
strands are annealed
all the way together and just resting
and then when you crank up the
temperature
in that reaction there's more and more
wiggling of the double helix and then
for any given stretch of dna there's a
temperature where there's so much
wiggling
that they separate and so then
you cool it down and they re-anneal so
that's kind of how pcr works
but it also has a statistical
interpretation
as you turn it up everything is wiggling
around
more and then as you cool it down
things converge to small ranges of
states near their ground state
and controlling the variance of a
distribution is useful in many
applications
i'm sure we can all think of many times
where there there's
a need for looseness or tightness or
flexibility

to tune a given model from loose to
tight
but what what what does that make you
think about
so um i just want to talk about
temperature like there's temperature in
this
um you know thermodynamics sense right
that controls
so much right about the energy of the of
the
particles and and all these things and i
just read like a really
interesting model and so i would have i
would have added the reference um i just
looked it up really quick so it's
a paper called dynamics of collective
action to conserve a large
common pool resource it's a paper that
just came out in
scientific reports by anderson at al but
in this paper they use this it's this
phi metric
and they use it to like talk about the
temperature
but it's not really like a temperature
but it's just like the
influence of all of the external
surroundings on
the behavior of the agents so i think
about in in the terms of active
inference
like what like the political arena the
news you know like all of these things
have influence on our our perception and
our action
right and so so can we have like a
temperature

like it like in terms of um having this
in in
uh in an active inference sense you know
and and they did
kind of a really nice job um baking that
into that
agent-based model is nice cool also
thinking about
action it reminds me of like a golf
swing or
chopping a knife the low temperature
action is the variance in the action is
very low
and so you might see that reflected by
like a groove that's very
entrenched but then the high variance
the higher temperature action is kind of
all over the place
and so we can imagine that especially
when we're learning
we don't know where the right place to
put the average
is so you don't want to have a tight
distribution too
early you want to start with a very high
temperature and a broad distribution
and then you want to hone in maybe like
there's a few ways that you might a few
successful end points you might get to
but ironically if you start there too
tightly then you'll fail
so that's kind of interesting to think
about
how about organisms that have to
minimize their free energy to stay alive
well alex writes that free energy
minimizing has been used to explain how
organisms

manage to keep themselves away from
thermodynamic equilibrium the ultimate
equilibrium
of dissipation with respect to the
external environment
in other words you have to maintain that
non-equilibrium steady state
because if you descend into total
equilibrium
it's equivalent to death and so
using a candle analogy you know the
flame is out
out out flame there's a point
where the system is no longer lit
the flame is out and then it's not
just uh that it's kind of it's like a
zero to one
with life and death even though we're
not going to the human
case or anything like that just taking
it thermodynamically with a candle
and then it's just like volume you can
just get more and more and more
dissipation of energy
from a candle to a flame to just a
complete inferno
more and more dissipation of energy and
then there's some value where you're not
dissipating energy as a system
you're dead well
let's map that on two organisms so
there's some point where the organism
dies not saying there's a moment where
it happens or a clean break
but there is a difference between
something that is no longer
actively maintaining a non-equilibrium
steady state versus something that isn't

and here is like a dog that's sick but
it's alive
and this is like a more vital and
healthy
dog and so the question would be
do species really minimize their free
energy
or do they just oh oh wait
that's a little bit more of a
generalization as long as the organism
is maintaining it's not equilibrium
steady state it's fair to say that it is
acting as if it's minimizing its free
energy
but then the question is maybe for other
levels of organization like a species
do higher levels of organization
actually minimize free energy in their
own right
or are they just more like a collection
of agents that are seeking to stay alive
and then the group stays alive as long
as you have individuals that are staying
alive
thermodynamically um so
i think that's partly where the mapping
is going with this whole
energy and information and life and
death but again
alex that'd be awesome spot you know
what is life what is death
these would be great things for you to
address any thoughts or we'll continue
all right so in the free energy
principle and
metaphysics section um
we pulled out a few quotes related to
where all of these predictive coding and

free energy principle
ideas come together it's a little bit
rehashing so we're not going to read
anything but
people should check out the paper and
yeah sometimes with these dense papers
we end up just
getting a lot on the slides just to get
it out there and unpack it and there's
many ways we could have probably
reordered stuff or
thought differently about it what about
this slide blue
okay
just that it's the just uh yeah i if
we're not going to read it we're not
going to read it the paper
was really good good good there let's
consider
the thought experiment that was done in
the paper which is this
sleeping and waking agent and so
um we're going to have these an agent
that has two different phases there's a
wake phase of the algorithm
um where the hidden caught where the
activity is driven bottom up by the
recognition weights so like when you're
awake
it's driven by the visual input coming
in
and then there's a sleep phase
where you can imagine the opposite is
true like the model updating is being
driven
by the generative model in the absence
of
being driven by stimuli and then

you take those two phases of the model
and combine them
into this r equation
so it's like you have night and day
light and wake and then the agent is
going to be modeling
that and i think dean you had a nice
take or interpretation on what this
sleep wake meant
what what what did this out what did
this example show or where did you see
it in the context of the paper
well sort of going back to this being an
example of this stochastic channel
so i i wondered whether or not
recognition is a
recognition which is then
consolidated is a two-way establishment
of what
sort of we've taken to be
what we now know versus what we don't so
is there
two parts to this being it's a two-way
establishment is there a build-up
in terms of the generative model as
probabilities are confirmed and
is there a build in is there a way to be
able to
to in that i guess it would be in the
sleep phase
or the non-weight phase is there a way
for us to be able to sort of purge the
error units
and then sort of discard them but again
i'm i'm saying this not knowing my
my negative f from my d sub k l
in terms of what this actually means
right but

but i do like the i do like the idea
that you can try to take a shot at it a
guess at it
and if you're completely off well fine
but it's not so
unreachable that you can't take a good
guess at it
thanks for really doing awesome on the
dot zero and bringing your perspective
because i totally agree with that and
like
i'm not even sure what the agent is
modeling here
it's like r of what let's find out
what this agent is doing and what it
means
but it is playing a role in the paper
which is using the model as just a
minimal
experiment to illustrate
the identity thesis which is the claim
that
the variational free energy so doing the
variational statistical inference
that model fitting is going to be
equal to the thermodynamic free energy
of the system so definitely alex
help us um understand what the model is
showing here
and also what maybe empirical system
could we actually look at
like can we collect data from a wake
person or a sleeping person or
another system to see if this is the
case
in equation 3.2 there's more unpacking
of the f in terms of a kl divergence and
then

um another term if you want to learn
more of the mathematical details
unpack that and work through it
in the discussion there
is some uh connection to the work
of lewis and others
how is the bayesian brain and the
position
of lewis uh
relating to folk psychology how do these
terms come together
section four later in section four
there's some formal logic um
ramsay sentences predicate logics
also it'd be very interesting to learn
about
what is being done here because we were
talking a lot about the mind and the
brain that sounded like
statistical i mean and as in
philosophical and then there's
things burning and thermodynamics and
then
here's a logical sentence that is
actually yeah blue what what did you see
here
so i i put a lot of this these little
comments into the slides because i was
like what is a ramsay sentence
and what are these like weird symbols
like i can do regular math but like when
you start to get into these like
i don't know tricky symbols it baffles
me always so a ramsey sentence is a
formal logic reconstruction
of a theoretical proposition that
attempts to draw a line between science
and metaphysics

which i didn't know that i've never
heard of these before um
and so i just couldn't really figure out
like and so this backwards yeah i was
trying to read this
like you know equal this equation here
down at the bottom and so
this backwards e is called the
existential quantifier
so like there exists uh x there exis
there exists of x
and then this upside down a is a
universal quantifier for all
so there exists an x for all y
and then the t i guess is the predicate
i don't know what that means so that y
uh and then this is a material by
conditional so i was really grateful for
the triple equal
unpacking because i had no idea what
that meant and then the identity
relationship so i couldn't even get
all the way through reading this so if
someone would read it to me that'd be
great
yep perhaps there exists x
for all y
and then the claim is of that or that
relationship
is that the predicate which it's the
part of a sentence
or a clause with a verb stating
something about the object like went
home
in john went home and then just to see
the three and the two
like the three is saying the material by
condition that's

the psychophysical part it's like saying
there's
you know something t of y and x those
are the parts that are
actually materially bi-conditional and
then there's an equal sign
and those are the identity or let's hear
it from alex read
in different languages another topic
that came up in this paper
as well as was in 16 is
what's the difference between a real
mind
brain system versus a
simulated mind brain system like
is every chess playing algorithm alive
just because it can play chess just
because people play chess
or does there have to be some other
aspect to the information dynamics
and it's something that both this paper
and again 16 with um wanja
weiss uh went in two
um any thoughts on real simulated
i really thought this was like the crux
of the paper
um so so that's that's my only thought
this identity thesis and the criterion
for distinguishing
you know a neural network versus an
actual neural network in my brain right
or a person with a with a neural network
or
an organism somehow
cool plankton or whatever it doesn't
have to be a brain but because because
alex did make that um distinction in the
paper awesome

so just a few more last notes for those
who stick around to the end
and listening to the good stuff first
how does the cell do more with less
it's kind of like asking how does the
brain do more with less
from a model fitting perspective if you
were to train the
um computer on tracing a baseball it
might use like
100 hamburgers worth of energy but then
you can trace the baseball
for a fraction of that one centi
hamburger so
that's the brain doing more with less
for model fitting
how does the cell do more than less more
with less in the biochemistry realm
and this helps us get an intuition for
how the brain might do more with less
in the information realm and it relates
to the hydrolysis
or the cutting with water of atp which
is a molecule with a high internal
energy
so you can use some of that internal
energy to do
other work and also release heat in
other words split atp
into adp and a free phosphate and extra
energy
free energy energy you can use and this
is what was
one of the most fascinating probably
facts that i learned
in my biochem education exactly how much
free energy is released
with a hydrolysis of atp and how is that

free energy used to do cellular work
the calculated delta g for the
hydrolysis of one mole of atp
is negative 7.3 k cal per mole so if you
just
burn the candle you get negative 7.3
units of energy that's under standard
conditions
in fact the delta g for the hydrolysis
in a living cell is almost double that
negative 14 k cows a mole how
is it vitalism is it magic well can't
rule that out
but there's a lot of tricks that can be
played with localizing
reactions so you can have a little tiny
bubble where the ph
is ridiculously acidic like one proton
vibrating around but it's so small that
the protons local acidity is unreal
and so there's all kinds of tricks that
can be played with
how things are combined and separated
and connected to each other and related
that help you get more energy out of
just the hydrolysis of an atp
so just think if you can get 2x
on burning the candle in a cell
probably you can do a lot more than 2x
on the information
in the environments is that physical
is it just statistical that's the
question
but it certainly is happening physically
with the cell so
it seems like it probably could be
happening in the brain as well
biochemistry blue no the cell has the

demon
right so so you know when we were
putting these slides together it's like
where's maxwell's demon in here like
the cell has the demon i guess i don't
know um
then where does active inference come
into play hashtag activelab
and so we searched for the term and it's
used
twice both times it's used
a little bit indirectly referring to the
literature
on active inference and the literature
on the fep
and also it's related to
topics we've talked about like top down
priors the umvelt
of the organism so it'd be super
interesting to hear
from alex where does active inference
come into play
in your agent as you had the thought
experiment or
just where does active inference come
into play
and then a key test and prediction
although
uh in another sort of classic three
sentence
paragraph alex does take it both ways
which is basically to ask
whether the identity thesis can be
tested
and uh he suggests that you could look
for a stochastic
code so in encoding just like we saw
earlier on with that sengupa

2013 paper the encoding that's going to
be like
the sort of trade-off between only
giving a little information but making
it the right info
so that the model can be updated
appropriately is to see if that is
empirically plausible and
this is an interesting philosophical
point
which is that the conjunction of the
identity thesis with
fep free energy principle is a
falsifiable hypothesis
even if the fep is not so in actin 14 15
20 we talked a lot about this what is
falsification
and relationship with the fep and so
it's almost like
fep is a carrier like it is deployed
in really clearly falsifiable ways
yet it is more of that carrier wave
that isn't directly falsifiable or i'm
not sure
where what he was getting at here but i
thought that was really fascinating
to think about not just oh it's a
corollary that's falsifiable
it's like we're taking the fep and
another
field and now that is going to be
something that we can both formalize and
falsify
but here we just have falsification here
we just had formalization and we had to
amalgamate
them in order to make something that
would take the best of both worlds

so let's hear from that and then this is
where i said he has it both ways
which is uh uh not that something is
true and false but actually
that the evidence is going to run
backwards to us he's saying
strong empirical evidence for the right
type of encoding
together with a formulation of cognitive
dynamics in terms of that variational
bayesian inference
supports the case so maybe there'd be so
much evidence
for some kind of encoding what kind of
encoding
how much evidence how would we know but
we'll find out
maybe it would be overwhelming positive
evidence
for the identity thesis and then as
always
then what what then then if then
all these cool learning terms that dean
has been helping us understand
what would happen then what would happen
if the identity thesis
is true or not and then what would we be
then so pretty fun
thank you both for participating any
last thoughts we'll have just a closing
section if anyone wants to ask a
question a live chat they can otherwise
we'll just
close it out can i ask one quick
question oh yes
yeah so i'm glad
blue brought up the thing that she
thought was really important about the

paper out
so i want to kind of swing all the way
back to the beginning
because i think identity as a as
proximal formalism
is really important and i part i want to
just read part of what he wrote
right directly under this section for a
transparent code because he never really
does get into this too much but
i think i think it kind of matters he
wrote stochastic coding and variational
inference in order for the thesis of
this paper
to make sense it must be kept in view
that the generative and recognition
densities of variational inference or
densities over and then he puts in
brackets possible
external causes of the sensory input
that is to say in the parlance of most
philosophers
and nearly all cognitive scientists they
are representations and it is their
representational function that defines
them as statistical models
fixing the encoding of the recognition
generative
densities allows us to directly relate
the representational work
in quotes done as the divergence between
the densities
is decreased decreased mind due to
physical work
that to me is would be amazing for him
to be able to
give that to me in in real terms of real
examples

of people doing that on a on a sort of
ad hoc basis because that would be
lovely to turn the what is um because i
understand appreciate the down the train
track thing
but the what if to be able to sort of
translate that what he wrote there in
terms of a transparent code
around what if would would make my day
and not just my model
it would make my model of my day
make my model of my day
but that sounds super interesting
and that part first about
representations we didn't even tackle
the representation question because also
in several other
streams we've asked about what is a
representation
internal representation where are these
representations what if they're
distributed in the environment
or among agents is that still a
representation but also that adjacent
possible
that's kind of like whoa if our
perception
is actually just the tip of the iceberg
of a really
deeper generative model and there's
cases where it seems
intuitive like oh i guess daytime is
contingent on having a model of night
or thinking a is contingent on
knowing that it could be a or not a but
even that whole framework for thinking
is still just part of a bigger process
and that's what i think a lot of

philosophy tries to grasp at
from a lot of sides
we're working with a kind of forceps but
there's like a bigger
picture well it sounds like we're also
working with something called a
transparent code
which blows me up because oh okay well
if that's all it was can you just tell
me
show me what that is and i know it's
going to be hard because it's
transparent but
hey let's go
blue i'm good
thanks guys well thanks to those who
watched
this live or in replay we hope that
you'll join us for the discussion
when it happens or after
thank you so great times talk to you
later dean talk